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# Enanthic acid

**Enanthic acid**, also called **heptanoic acid**, is an organic compound composed of a seven-carbon chain terminating in a carboxylic acid functional group. It is a colorless oily liquid with an unpleasant, rancid odor.[1] It contributes to the odor of some rancid oils. It is slightly soluble in water, but very soluble in ethanol and ether. Salts and esters of enanthic acid are called **enanthates** or **heptanoates**.

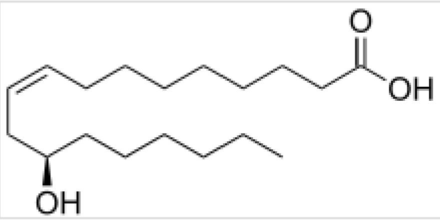
Its name derives from the Latin *oenanthe* which is in turn derived from the Ancient Greek *oinos* "wine" and *anthos* "blossom."

**Enanthic acid[1]**


| Names | |
|---|---|
| Preferred IUPAC name | Heptanoic acid |
| Other names | Enthanoic acid; Enthanylic acid; Heptoic acid; Heptylic acid; Oenanthic acid; Oenanthylic acid; 1-Hexanecarboxylic acid; C7:0 (lipid numbers) |

| Identifiers | |
|---|---|
| CAS Number | 111-14-8 (https://commonchemistry.cas.org/detail?cas_rn=111-14-8) ✓ |
| 3D model (JSmol) | Interactive image (https://chemapps.stolaf.edu/jmol/jmol.php?model=O%3DC%28O%29CCCCCC) |
| ChEBI | CHEBI:45571 (https://www.ebi.ac.uk/chebi/searchId.do?chebiId=45571) ✓ |
| ChEMBL | ChEMBL320358 (https://www.ebi.ac.uk/chembldb/index.php/compound/inspect/ChEMBL320358) ✓ |
| ChemSpider | 7803 (https://www.chemspider.com/Chemical-Structure.7803.html) ✓ |

## Production




Ricinoleic acid, a fatty acid obtained from castor bean oil, also occurs as its methyl ester, methyl ricinoleate, which is the main precursor to enanthic acid.

The methyl ester of ricinoleic acid, obtained from castor bean oil, is the main commercial precursor to enanthic acid. It is pyrolyzed to the methyl ester of 10-undecenoic acid and heptanal, which is then air-oxidized to the carboxylic acid. Approximately 20,000 tons were consumed in Europe and US in 1980.[2]

Laboratory preparations of enanthic acid include permanganate oxidation of heptanal[3] and 1-octene.[4]

## Uses

Enanthic acid is used in the preparation of esters, such as ethyl enanthate, which are used in fragrances and as artificial flavors. Enanthic acid is used to esterify steroids in the preparation of drugs such as testosterone enanthate, trenbolone enanthate, drostanolone enanthate, and methenolone enanthate (Primobolan).

The triglyceride ester of enanthic acid is the triheptanoin, which is used in certain medical conditions as a nutritional supplement.

## Safety

Enanthic acid is toxic if swallowed and corrosive.[5]

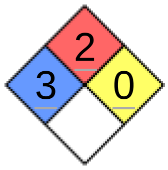
## See also

- List of saturated fatty acids
- List of carboxylic acids

## References


1. *Merck Index*, 11th Edition, **4581**
2. David J. Anneken, Sabine Both, Ralf Christoph, Georg Fieg, Udo Steinberner, Alfred Westfechtel "Fatty Acids" in Ullmann's Encyclopedia of Industrial Chemistry 2006, Wiley-VCH, Weinheim. doi:10.1002/14356007.a10_245.pub2 (https://doi.org/10.1002%2F14356007.a10_245.pub2)
3. John R. Ruhoff (1936). "N-Heptanoic Acid". *Organic Syntheses*. **16**: 39. doi:10.15227/orgsyn.016.0039 (https://doi.org/10.15227%2Forgsyn.016.0039).
4. Donald G. Lee, Shannon E. Lamb, Victor S. Chang (1981). "Carboxylic Acids from the Oxidation of Terminal Alkenes by Permanganate: Nonadecanoic Acid". *Organic Syntheses*. **60**: 11. doi:10.15227/orgsyn.060.0011 (https://doi.org/10.15227%2Forgsyn.060.0011).
5. "Heptanoic Acid - Pubchem" (https://pubchem.ncbi.nlm.nih.gov/compound/Heptanoic-acid).


| | |
|---|---|
| DrugBank | DB02938 (https://www.drugbank.ca/drugs/DB02938) ✓ |
| ECHA InfoCard | 100.003.490 (https://echa.europa.eu/substance-information/-/substanceinfo/100.003.490) |
| KEGG | C17714 (https://www.kegg.jp/entry/C17714) ✓ |
| PubChem CID | 8094 (https://pubchem.ncbi.nlm.nih.gov/compound/8094) |
| UNII | THE3YNP39D (https://precision.fda.gov/uniisearch/srs/unii/THE3YNP39D) ✓ |
| CompTox Dashboard (EPA) | DTXSID2021600 (https://comptox.epa.gov/dashboard/chemical/details/DTXSID2021600) |
| InChI | InChI=1S/C7H14O2/c1-2-3-4-5-6-7(8)9/h2-6H2,1H3,(H,8,9) ✓<br>Key: MNWFXJYAOYHMED-UHFFFAOYSA-N ✓<br><br>InChI=1/C7H14O2/c1-2-3-4-5-6-7(8)9/h2-6H2,1H3,(H,8,9)<br>Key: MNWFXJYAOYHMED-UHFFFAOYAP |
| SMILES | O=C(O)CCCCCC |
| **Properties** | |
| Chemical formula | $C_7H_{14}O_2$ |
| Molar mass | 130.187 g·mol$^{-1}$ |
| Appearance | colorless oily liquid |
| Density | 0.9181 g/cm$^3$ (20 °C) |
| Melting point | −7.5 °C (18.5 °F; 265.6 K) |

| | |
|---|---|
| Boiling point | 223 °C (433 °F; 496 K) |
| Solubility in water | 0.2419 g/100 mL (15 °C) |
| Magnetic susceptibility (χ) | $-88.60 \cdot 10^{-6}$ $cm^3$/mol |
| **Hazards** | |
| **NFPA 704** (fire diamond) |  |
| **Lethal dose** or concentration (LD, LC): | |
| $LD_{50}$ (median dose) | 6400 mg/kg (oral, rat) |
| **Related compounds** | |
| Related compounds | Hexanoic acid, Octanoic acid |
| Except where otherwise noted, data are given for materials in their standard state (at 25 °C [77 °F], 100 kPa). | |
| ✓ verify (what is ✓✗ ?) | |
| Infobox references | |